

MOTIVATION  Target group: Politically interested citizens and political journalists. Need: Many feel Finnish politics has become more polarised. The best evidence from parliamentary speech ends in 2018 (Simola et al. 2025). We bring it up to date. Questions - How different do the parties sound in parliament, and which ones sound alike? - Are parties moving apart, and are MPs within a party sounding more alike? - Which words and topics set the parties apart? Benefit: An interactive picture of how parties have talked in parliament 2015–2026, built on public data and standard methods.	DATA COLLECTION  Eduskunta open data API: all plenary speeches 2015–2026 with speaker, party and date. Public, CC BY 4.0, available now. Yle election compass 2023: about 2,000 candidates' answers by party, anonymised, CC-BY. Used to check our speech-based map against what parties told voters. Data management: one speech table, one MP table; raw data kept as is, derived data versioned in the team repo. Stretch: we have asked Yle for the compass data with MP names. If it arrives in September, we can also compare individual MPs' stated and spoken positions.	PREPROCESSING  Goal: text the model can classify only by content, not by clues about who is speaking. - Keep Finnish-language speeches; drop procedural turns. - Remove party and MP names from the text. - Lemmatise with standard Finnish NLP tools; turn speeches into word counts. - Combine each MP's speeches per term for the maps. - Handle party renames and MPs who switch groups. Avoiding misleading trends - Split train and test data by MP, so the model learns party views, not one person's style. - Same number of speeches per party and term, because more data alone makes parties look more different (Gentzkow et al. 2019).	EXPLORATORY DATA ANALYSIS (EDA)  - Speeches and speakers per party per term. - Topics per party: different topics, or the same topics discussed differently? - Government vs. opposition changes across the Sipilä, Marin and Orpo terms. - Most distinctive words per party. - Baseline: how well can party be guessed from text at all?	VISUALIZATIONS  1. Party map per term: parties placed close together if the model often mixes them up; MPs as dots inside their party. For 2023, shown next to the map from election compass answers. 2. Change over time: distance between parties and spread within parties, 2015–2026. 3. What sets them apart: the words that best identify each party. Interactive: slider over terms; hover for details. Demo: paste a speech and see which party the model guesses, and why.
LEARNING TASK  Supervised classification, following Gentzkow et al. (2019) and Simola et al. (2025). Task: predict the speaker's party from speech text, separately for each term. Control: predict government vs. opposition, to see how much is about being in government rather than party. Polarisation = how easily the model tells parties apart, and which ones it mixes up. Diversity = how much MPs within a party differ from each other in the model's view. Stretch: predict each MP's compass answers from speech; compare stated and spoken positions.	LEARNING APPROACH  Models: simple linear text classifiers, because their word weights can be read and shown. Stretch: a Finnish BERT model, only if the simple model works by early October. Evaluation: standard classification metrics per term; confusion matrices; confidence intervals from resampling MPs. Validation: cross-validation by MP; party distances from speech checked against party distances from compass answers. Compared with: Simola et al. (2025) for 1907–2018; Gronow & Malkamäki (2024) for social media 2015–2023.		COMMUNICATION OF RESULTS  Blog post: plain language, three main findings, interactive charts. Demo: "which party said this?" (if time allows). Technical report: methods, validation, what we learned (5 pages). Spotlight line <i>"A parliament can change in two ways: parties move apart, or they stop disagreeing inside. We measured both, for every party, 2015–2026."</i>	DATA PRIVACY AND ETHICAL CONSIDERATIONS  - All data is public record. No consent needed; no personal data beyond MPs' public roles. - Model results are estimates, not facts. The blog reports by party; individual MPs only in the technical report; no rankings of MPs. - The compass file is anonymised on purpose; we use it at party level only and do not try to identify candidates. - Small parties have less data and are shown as less certain. Swedish-language speeches are left out, and we say so.
RISKS - Parties may differ mostly in what they talk about, not how. We show this in the EDA and say it with the results. - Changes may be small or absent. "No clear change" is a fine result. - Being in government changes how MPs speak; the control task shows how much.	ADDED VALUE  - First analysis of party differences in Finnish parliamentary speech for 2019–2026. - Model guesses become useful through the maps: how often two parties are mixed up becomes a distance readers can see and follow over time. - Re-runnable each year; works for any parliament with similar data.			LEGEND WEEK 1: Data collection/preprocessing WEEK 2: EDA & visualizations WEEKS 3-4: Machine/deep learning WEEK 5: Fairness & data privacy